

## WEEKLY REPORT

**Dear Zeina Ahmad**

In the below Table you will see a specified table that includes the good, the bad & the improvements that can be done on your Classification Task

| Correctly Done                                                                                                                                                                 | Can Be Improved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>Encoding</b></li> <li>• <b>Handle Outlier</b></li> <li>• <b>Handle Imbalanced</b></li> <li>• <b>Applying Algorithms</b></li> </ul> | <ul style="list-style-type: none"> <li>• <b>Scaling</b></li> <li>• <b>Handle multicollinearity using drop cols iteratively with high vif or pca (not necessary to increase performance, the integrity will be)</b></li> <li>• <b>You can apply other techniques for outlier as transformation (log), capping, fill by median (try and determine the good one)</b></li> <li>• <b>You can try encoding using one-hot, label or manual encoder as type1:1, type2:2,...</b></li> <li>• <b>You can apply robust algorithm as random forest, xgboost,...</b></li> <li>• <b>You can apply cross-validation to know genera performance of model on all data</b></li> <li>• <b>Split data into train and test before any preprocessing</b></li> <li>• <b>You can add more features as total_members, total_night, percent_can((p_c)/(p_c + p_not_c</b></li> </ul> |

